

Carlos Moreno Rojas

Biomedical & Electrical Engineer | R&D | Embedded Systems | Instrumentation

Santiago, Chile | camoro2002@uc.cl | LinkedIn | Portfolio

PROFESSIONAL PROFILE

Biomedical and electrical engineer and M.Sc. candidate with hands-on experience in **biomedical instrumentation, embedded systems, rapid prototyping, hardware-software integration, computer vision, and medical-device development**. Experienced in translating technical requirements into functional prototypes through iterative design, system integration, experimental validation, and multidisciplinary engineering.

EDUCATION

Pontificia Universidad Católica de Chile <i>M.Sc. Candidate in Biomedical Engineering – Biomedical Instrumentation</i>	<i>2026–2027 (expected)</i> Santiago, Chile
Pontificia Universidad Católica de Chile <i>Professional Degree in Electrical Engineering</i>	<i>2025–2026</i> Santiago, Chile
Pontificia Universidad Católica de Chile <i>Bachelor's Degree in Engineering Sciences – Major in Biomedical Engineering</i>	<i>2021–2024</i> Santiago, Chile

ENGINEERING AND R&D EXPERIENCE

Candel Medical Company – Omar Espinoza <i>Hardware/Firmware Engineering Intern – Transcranial Electrical Stimulation (tES)</i>	<i>Jan. 2026–Mar. 2026</i> Santiago, Chile
- Improved a medical-device prototype toward product-level integration through circuit redesign, PCB debugging, firmware development, and experimental validation. - Worked with ESP32 and STM32, BLE communication, signal generation and synchronization, digital control, power optimization, and hardware-software integration for tDCS/tACS/tRNS stimulation modes.	

Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo & Prof. Domingo Mery <i>Biomedical Instrumentation / Master's Research</i>	<i>2026–Present</i> Santiago, Chile
- Designed and developed a custom ex vivo rat liver perfusion platform, integrating pressure, flow, temperature, oxygenation, pulse damping, bubble management, thermal control, and fluidic subsystems. - Developed a modular hydrodynamic monitor and data-acquisition system using clinical sensors, embedded electronics, and real-time software; calibrated and validated measurement channels against reference instrumentation including a Fluke ProSim 8. - Currently developing an ICG-NIR imaging methodology for spatial and temporal fluorescence analysis using camera characterization, optical filtering, controlled acquisition, and quantitative image processing.	

Ciervo UC – CYBATHLON 2024 <i>Electrical Area Leader – Robotic Transfemoral Prosthesis</i>	<i>2024</i> Santiago, Chile / Zürich, Switzerland
Led electrical/electronic development of an EMG-controlled robotic transfemoral prosthesis, integrating electronics, control, and mechanical subsystems through iterative prototyping and system-level troubleshooting for CYBATHLON 2024 at ETH Zürich.	

Meldic Ltda. <i>Electrical Engineer – FPGA-Based Upgrade for Universal Testing Machine</i>	<i>2025–2026</i> Santiago, Chile
Developed an embedded processing architecture using Zybo Z7, Vivado, and VHDL for force, displacement, deformation, and speed-control signals in mechanical-testing equipment.	

SELECTED ENGINEERING AND BIOMEDICAL PROJECTS

Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo <i>Organ Preservation and Supercooling – Borealis UC</i>	<i>2024</i> Santiago, Chile
Co-developed from concept a supercooling-based organ preservation system, integrating mechanical design, electronics, thermal control, fluid handling, miniaturization, and experimental testing.	

Medical Image Segmentation with U-Net <i>Pontificia Universidad Católica de Chile – Dr. Marcelo Andia</i>	<i>2025</i> Santiago, Chile
Developed a U-Net-based neural network for lung segmentation in CT and MRI, applying deep learning to biomedical image analysis.	

Institute for Biological and Medical Engineering (IIBM UC) – Dr. Martín Miranda
3D Design for GE Vscan Air Accessory

2023

Santiago, Chile

- Designed a CAD-based cover/enclosure for the **GE Vscan Air** handheld ultrasound transducer using rapid prototyping and 3D design.

LEADERSHIP AND ADDITIONAL EXPERIENCE

Biomedical Engineering Student Chapter PUC (CEIBM)
President

2024–2025

Santiago, Chile

IEEE EMBS PUC Chile
Webmaster

2024

Santiago, Chile

Faculty of Physics, Pontificia Universidad Católica de Chile
Laboratory Teaching Assistant – FIS154

Mar.–Jun. 2024

Santiago, Chile

TECHNICAL SKILLS

Embedded / Electronics: Analog and digital electronics, ESP32, STM32, Arduino, clinical sensors, PCB design/debugging, BLE, serial communication, instrumentation, and hardware-software integration.

Prototyping / System Integration: Rapid prototyping, iterative design, design-build-test cycles, electromechanical integration, sensor integration, system-level troubleshooting, proof-of-concept development, CAD, and 3D printing.

FPGA / Digital Design: Vivado, VHDL, Zybo Z7, FPGA-based digital systems, and embedded signal-processing architectures.

Software / Data: Python, C/C++, data acquisition, signal processing, data analysis, and scientific computing.

Imaging / Machine Learning: Advanced image processing, computer vision, pattern recognition, image segmentation, deep learning, neural networks, and quantitative image analysis.

Biomedical / Optics: Medical-device prototyping, physiological monitoring, biosensors, NIR/fluorescence imaging, optical filters, camera characterization, validation, and calibration.

Leadership / Collaboration: Technical leadership, interdisciplinary teamwork, project coordination, technical communication, teaching and mentoring, and collaborative problem-solving.

LANGUAGES

Spanish: Native

English: TOEIC 900/990 – B2 / ALTE 3